

Category	Item	Value	Unit
Material	Concrete	100	m³
	Steel	50	kg
	Brick	200	kg
	Wood	30	m³
Labor	Construction Worker	10	hr
	Electrician	5	hr
	Plumber	3	hr
	Painter	2	hr
Equipment	Excavator	1	hr
	Truck	2	hr
	Generator	1	hr
	Toolbox	1	hr

--	--	--	--	--

--	--	--	--

9702/31

May/June 2013

2 hours

Additional Materials: As listed in the Confidential Instructions.

DO **NOT** WRITE IN ANY BARCODES.

You are reminded of the need for good English and clear presentation in your answers.

You may lose marks if you do not show your working or if you do not use appropriate units.

Additional answer paper and graph paper should be used only if it becomes necessary to do so.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
Total	

This document consists of **12** printed pages.

You may not need to use all of the materials provided.

For
Examiner's
Use

- 1 In this experiment, you will investigate a system in equilibrium due to several turning forces.

- (a) Measure and record the distance L between the two holes in the wooden strip as shown in Fig. 1.1.

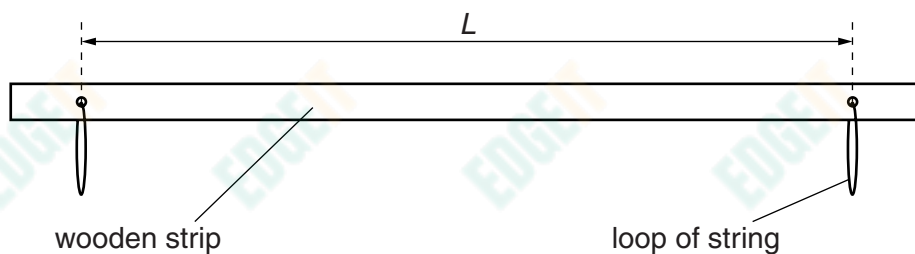


Fig. 1.1

$L = \dots\dots\dots$ m [1]

- (b) Write down the mass M given on the card.

$M = \dots\dots\dots$ kg

- (c) (i) Set up the apparatus as shown in Fig. 1.2, with mass $m = 0.040$ kg.

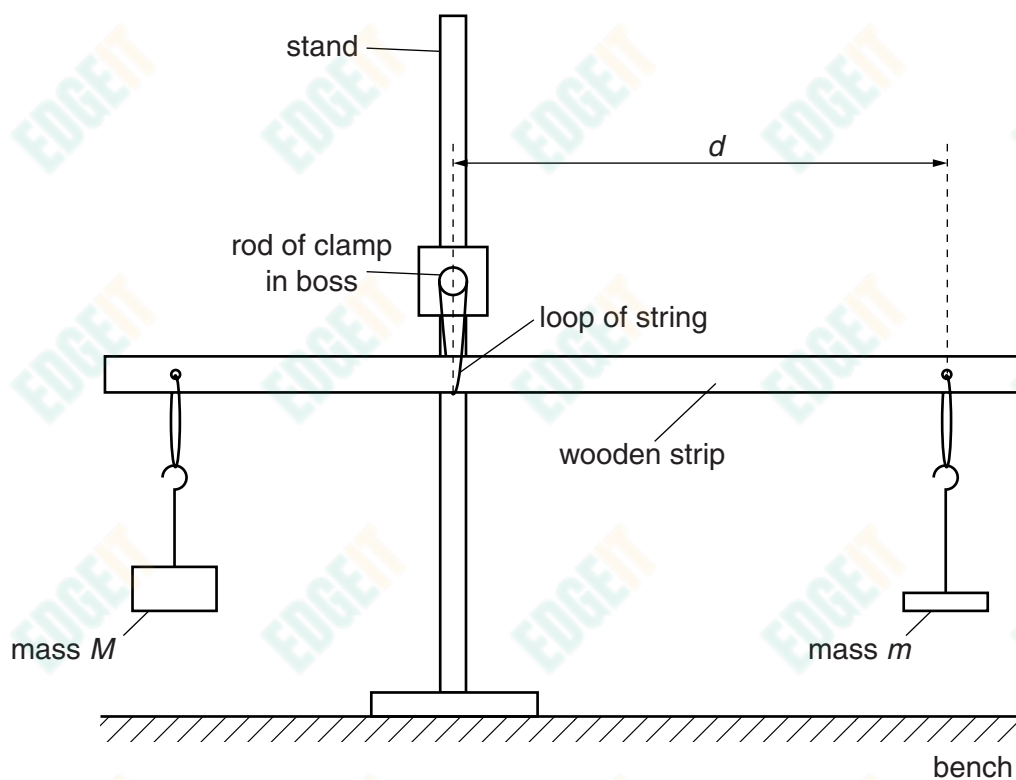


Fig. 1.2

- (ii) Adjust the position of the wooden strip until it balances.

Measure and record the distance d , as shown in Fig. 1.2.

$d = \dots\dots\dots$ m [1]

For
Examiner's
Use

(d) Vary m and repeat (c)(ii) until you have six sets of readings of m and d .

Include values of $\frac{1}{d}$ in your table.

For
Examiner's
Use

☐

[10]

(e) (i) Plot a graph of $\frac{1}{d}$ on the y -axis against m on the x -axis.

[3]

(ii) Draw the straight line of best fit.

[1]

(iii) Determine the gradient and y -intercept of this line.

gradient =

y -intercept =

[2]

For
Examiner's
Use

- (f) The quantities d and m are related by the equation

$$\frac{1}{d} = Pm + Q$$

where P and Q are constants.

Using your answers in (e)(iii), determine the values of P and Q .
Give appropriate units.

$$P = \dots\dots\dots$$

$$Q = \dots\dots\dots$$

[1]

- (g) The constant P is related to L and M by

$$P = \frac{1}{kML}$$

where k is a constant.

Using your answers in (a), (b) and (f), calculate a value for k .
You need not include units for k .

$$k = \dots\dots\dots$$

[1]

You may not need to use all of the materials provided.

For
Examiner's
Use

2 In this experiment, you will investigate the motion of a wooden rod supported by a string.

(a) (i) Set up the apparatus as shown in Fig. 2.1 with height h approximately 40 cm.

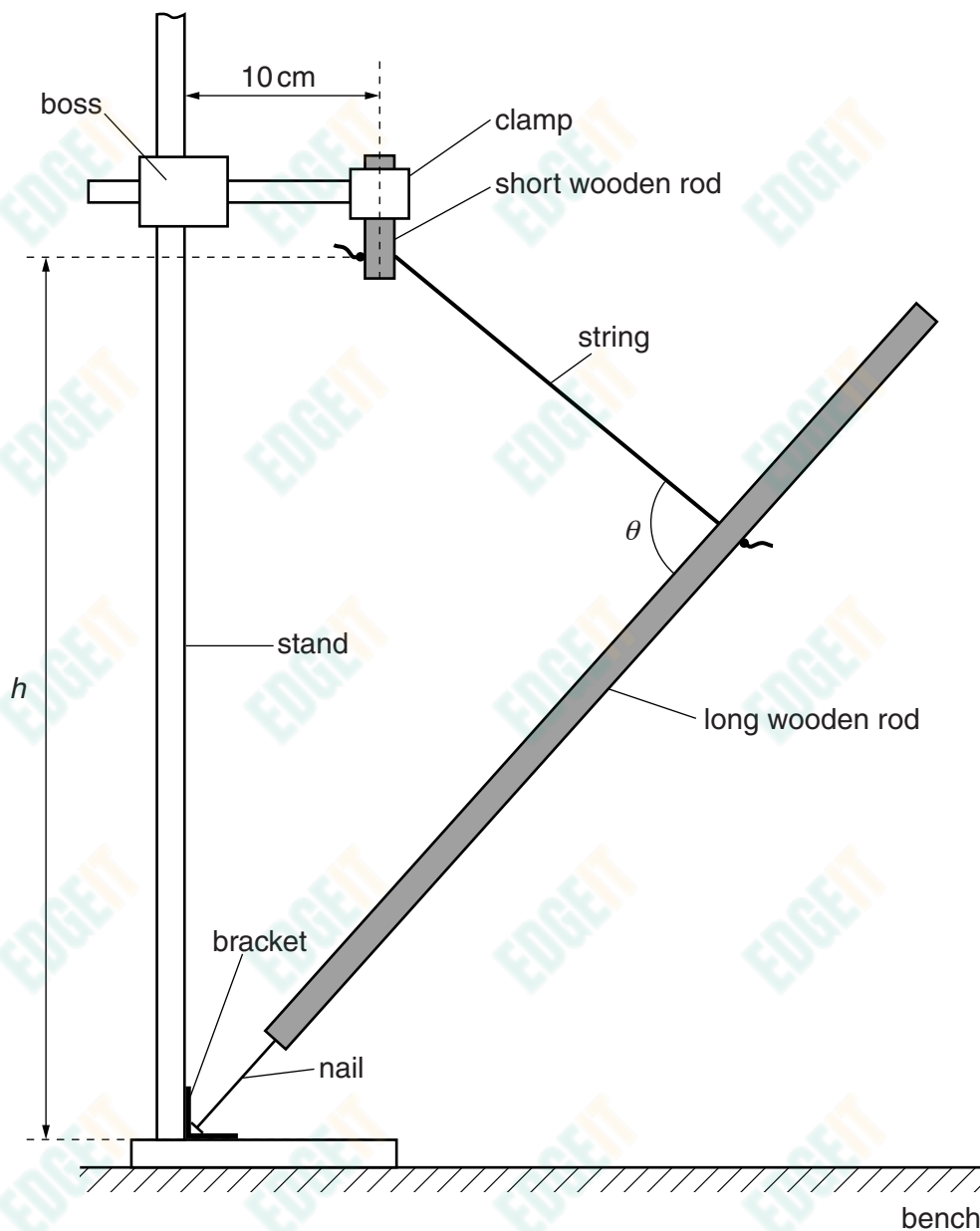


Fig. 2.1

The short wooden rod should be held firmly in the clamp.

Place the bracket on the base of the stand and rest the head of the nail in the centre of the bracket.

- (ii) Measure and record the angle θ between the string and the long wooden rod as shown in Fig. 2.1.

For
Examiner's
Use

$\theta = \dots\dots\dots$ [2]

- (iii) Estimate the percentage uncertainty in your value of θ .

percentage uncertainty = $\dots\dots\dots$ [1]

--

- (iv) Calculate $\sin \theta$.

$\sin \theta = \dots\dots\dots$ [1]

--

- (b) Gently displace the end of the long wooden rod to the left as shown in Fig. 2.2 (top view).

For
Examiner's
Use

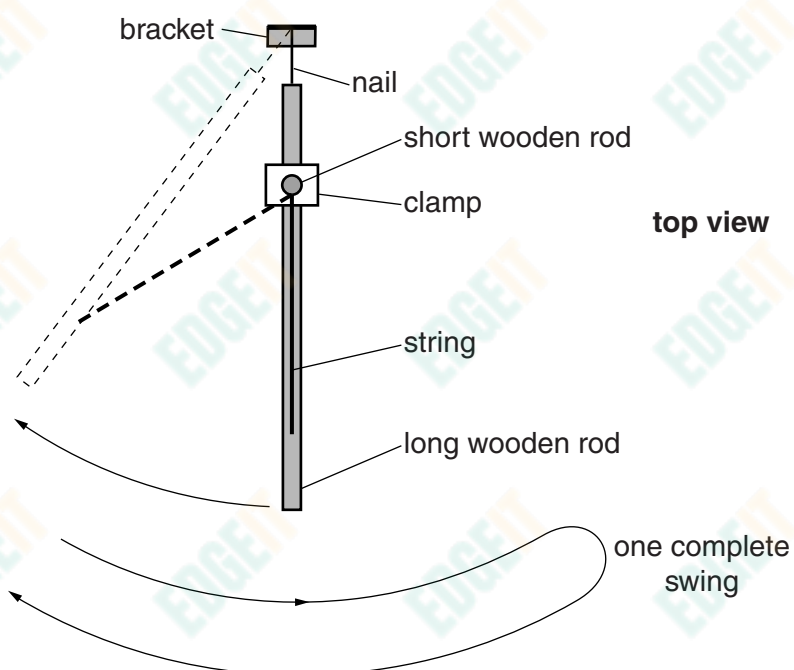


Fig. 2.2

Release the rod and watch the movement.
The rod will move to the right and back towards the left, completing a swing.
The time taken for one complete swing is T .
By timing several of these complete swings, determine an accurate value for T .

$T = \dots\dots\dots$ [2]

(c) (i) Reduce h to approximately 20 cm.

(ii) Repeat (a)(ii), (a)(iv) and (b).

For
Examiner's
Use

$\theta =$

$\sin \theta =$

$T =$ [3]

- (d) It is suggested that the relationship between T and θ is

$$T^2 = k \sin \theta$$

where k is a constant.

- (i) Using your data, calculate two values of k .

first value of $k =$

second value of $k =$ [1]

- (ii) Justify the number of significant figures that you have given for your values of k .

.....

 [1]

- (iii) Explain whether your results in (d)(i) support the suggested relationship.

.....

 [1]

For
Examiner's
Use

- (e) (i) Describe four sources of uncertainty or limitations of the procedure for this experiment.

For
Examiner's
Use

1.

2.

3.

4.

[4]

- (ii) Describe four improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

1.

2.

3.

4.

[4]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

University of Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.